

Chih-Hui (John) Ho

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Google Scholar

Education

- Sep. 2019 - **University of California San Diego, La Jolla, CA.**
Mar. 2024 PhD in Electrical Computer Engineering
Advisor: Professor Nuno Vasconcelos
- Sep. 2017 - **University of California San Diego, La Jolla, CA.**
Jun. 2019 M.S. in Computer Science, GPA: 3.87/4.0
- Sep. 2012 - **National Chiao Tung University, Hsinchu, Taiwan.**
Jun. 2016 B.S. in EECS Honor Program, GPA: 4.15/4.3

Professional Experience

- Present **Senior Applied AI Researcher, Articul8.ai (Intel AI Spinout).**
- Apr. 2024 - **Applied AI Researcher.**
- Dec. 2024
- Key contributor to end-to-end productionization of domain- and task-specific LLM and VLM systems (e.g., text-to-SQL, agentic tool calls, Verilog, question-to-image retrieval, single-line diagram and CAD understanding), owning data sourcing, training, evaluation, and model serving, and delivering multiple production models that outperformed open-source baselines on domain benchmarks for more than 15% accuracy.[Verilog Blog, Benchmark Blog, and Energy Model Blog].
 - Designed scalable multimodal document understanding and RAG systems (OCR, layout analysis, table parsing, diagrams), converting research prototypes into production pipelines processing millions of documents and delivering over 25% speed improvement over the initial design. [Blog for Table Agent on AWS]
 - Built and trained LLMs and VLMs up to 70B parameters using 200+ GPUs, covering continued pretraining (CPT) and post-training (SFT, RL), with production serving via vLLM and LitServe on NVIDIA GPUs and Intel Gaudi.
 - Drove technical strategy across high-ambiguity, aligning research, engineering, and product; mentored researchers and interns, and authored NeurIPS [Paper, Blog] and ICCV [Website] publications, translating research into production impact.
- Jan. 2018 - **Graduate Student Researcher, Statistical Visual Computing Lab, UCSD.**
- Mar. 2024
- Worked on large foundational models, self-supervised learning, defect detection, and long-tail recognition
 - Collaborated with Korean Polytechnic University and PCB/Tire company for defect detection research
- Jun. 2023 - **Mitsubishi Electric Research Lab (MERL) Research Intern.**
- Sept. 2023
- Conducted research on industrial anomaly detection using visual language foundational model
- Jun. 2021 - **Amazon AWS Applied Scientist Intern.**
- Sept. 2021
- Developed a visual grounding transformer model with 1.3x smaller size and 3x faster speed
 - Published a paper in ECCV2022 Workshop
- Sep. 2017 - **Research Volunteer, San Diego Supercomputer Center.**
- Dec. 2017
- Reduced error of large scale operational facility data (200 GB) in scientific workflow by 23%
- Nov. 2016 - **Research Assistant, NCTU Computer Vision Research Center.**
- Jun. 2017
- Developed deep learning model for human activity analysis in aerial images
 - Designed bill serial number recognition system with more than 99 % accuracy
 - Developed camera calibration algorithm and implemented the algorithm into prototype
 - Developed algorithm for automated optical inspection (AOI) for bobbin defects

- Jan. - Dec. **Software Engineer Internship, Industrial Technology Research Institute.**
 2015 o Developed a prototype to calibrate robotic arm with an industrial camera
 o Represented ITRI to attend 2015 Taiwan Automation Intelligence and Robot Show
 o Received Mechanical and Systems Research Lab Prospective Project Excellence Award
- Jul. - Aug. **Research Internship, Cornell University Advanced Multimedia Lab.**
 2014 Design algorithm to generate image collage based on emotional ROIs

Publication and Patent

- 2025 **Chih-Hui Ho**, Giovanna Marinho, Felipe Viana, Varad Pimpalkhute, Rodolfo Tonoli, Andre Von Zuben, "DocQIR-Emb: Document Image Retrieval with Multi-lingual Question Query", In *Neural Information Processing Systems (NeurIPS) UniReps Workshop*, 2025 [Paper, Blog]
- 2024 **Chih-Hui Ho**, Kuan-Chuan Peng, Nuno Vasconcelos, "Long-Tailed Anomaly Detection with Learnable Class Names", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- 2024 **Chih-Hui Ho**, and Kuan-Chuan Peng, "Long-Tailed Anomaly Detection in Images", US Patent Application 18/590210, Feb. 28, 2024.
- 2024 Simeng Zheng, **Chih-Hui Ho**, Wenyu Peng, Paul H Siegel, "Flash-Gen: Spatio-temporal generator for flash memory systems", In *IEEE Transactions on Communications*, 2024
- 2024 **Chih-Hui Ho***, Tz-Ying Wu*, Nuno Vasconcelos, "ProTeCt: Prompt Tuning for Taxonomic Open Set Classification", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- 2023 **Chih-Hui Ho**, SungBal Seo, NaYeon Kim, Pin-Ying Wu, YouSuk Bae, Nuno Vasconcelos, "Unsupervised PCB Anomaly Segmentation with Foundational Model", *Electronic Imaging (EI), Intelligent Robotics and Industrial Applications using Computer Vision*, **Oral**, 2024.
- 2023 **Chih-Hui Ho***, Yuwei Zhang*, Nuno Vasconcelos, "Toward Unsupervised Realistic Visual Question Answering", In *International Conference on Computer Vision (ICCV)*, 2023, [Paper]
- 2023 Simeng Zheng, **Chih-Hui Ho**, Paul H Siegel, "Modeling Flash Memory Channels Using Conditional Generative Nets", *Design Automation and Test in Europe*, 2023, [Paper]
- 2023 **Chih-Hui Ho**, Ziheng Huang, NaYeon Kim, YouSuk Bae, Nuno Vasconcelos, "Tire Defect Detection with Limited Annotation", *Electronic Imaging (EI), Intelligent Robotics and Industrial Applications using Computer Vision*, **Oral**, 2023. [Paper]
- 2022 **Chih-Hui Ho**, Nuno Vasconcelos, "DISCO: Adversarial Defense with Local Implicit Functions", In *Neural Information Processing Systems (NeurIPS)*, 2022 [Paper]
- 2022 **Chih-Hui Ho**, Srikar Appalaraju, Bhavan Jasani, R. Manmatha, Nuno Vasconcelos, "YORO - Lightweight End to End Visual Grounding", In *European Conference On Computer Vision Workshop (ECCVW)*, 2022 [Paper] [Blog]
- 2022 Brandon Leung, **Chih-Hui Ho**, Nuno Vasconcelos, "Black-Box Test-Time Shape REFINement for Single View 3D Reconstruction", In *IEEE Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)*, 2022 [Paper]
- 2020 **Chih-Hui Ho**, Nuno Vasconcelos, "Contrastive Learning with Adversarial Examples", In *Neural Information Processing Systems (NeurIPS)*, 2020
- 2020 Tz-Ying Wu, Pedro Morgado, Pei Wang, **Chih-Hui Ho**, Nuno Vasconcelos, "Solving Long-tailed Recognition with Deep Realistic Taxonomic Classifier", In *European Conference on Computer Vision (ECCV)*, 2020
- 2020 **Chih-Hui Ho**, Bo Liu, Tz-Ying Wu, Nuno Vasconcelos, "Exploit Clues from Views: Self-Supervised and Regularized Learning for Multiview Object Recognition", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020
- 2020 Jen-Hui Chuang, **Chih-Hui Ho**, Ardian Umam, HsinYi Chen, Mu-Tien Lu, Jenq-Neng Hwang, Tai-An Chen, "Geometry-Based Camera Calibration Using Closed-Form Solution of Principal Line", *IEEE Transaction on Image Processing (TIP)* 2020. [Paper] [ArXiv]
- 2019 **Chih-Hui Ho**, Pedro Morgado, Amir Persekian, Nuno Vasconcelos, "PIEs: Pose Invariant Embeddings", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019
- 2019 **Chih-Hui Ho***, Brandon Leung*, Erik Sandstrom, Yen Chang, Nuno Vasconcelos, "Catastrophic Child's Play: Easy to Perform, Hard to Defend Adversarial Attacks", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019

- 2019 Brandon Leung, **Chih-Hui Ho**, Amir Persekian, David Orozco, Yen Chang, Erik Sandstrom, Bo Liu, Nuno Vasconcelos, "OOWL500: Overcoming Dataset Collection Bias in the Wild", *[ArXiv]*
- 2018 Yu-Shiuan Tsai, Yi-Yu Hsieh, **Chih-Hui Ho**, Ya-Ching Chang, Yao-Yuan Chang, Heng-Jyun Lin, Han-Yang, Wang; Yu-Chen Chou, Jen-Hui Chuang, "Rule-Based Optical Character Recognition for Serial Number on Renminbi Banknote", In *IS&T Electronic Imaging 2018 (EI)* (**oral** presentation)

Academic Services

- Reviewer NeurIPS, CVPR(**Outstanding Reviewer** for 2021), ECCV, ICCV, TPAMI, ICML (**Outstanding Reviewer** for 2022), ICLR, ACCV, WACV, CVPR Learning with Limited Labelled Data for Image and Video Understanding Workshop, ECCV Imbalance Problems in Computer Vision Workshop, ICIP
- Program Committee ECCV2020 Imbalance Problems in Computer Vision Workshop, CVPR2022 Learning with Limited Labelled Data for Image and Video Understanding Workshop
- Volunteer CVPR 2020 Area Chair Meeting, San Diego

Teaching Experience

- UCSD ECE 175A Elements of Machine Intelligence [Discussion Lead](Recording), ECE 271A Statistical Learning I, ECE 271B Statistical Learning II, ECE 271C Deep Learning and Applications, CSE166 Image Processing

Selected Projects

- Jan. 2021 - **Defect Detection**, *Pytorch*.
 Mar. 2024
 - o Collaborating with Korea Polytechnic University to detect tire and pcb defects
 - o 2 paper published in EI conference about defect detection using self-supervised learning techniques and segment anything model (SAM)
- Jan. - Mar. 2018 **2018 Kaggle data science bowl**, *Keras*.
 2018
 - o Implemented image segmentation deep learning models for medical images
 - o Ranked top 18% in the competition
- Nov. 2016 - **Deep learning based human activity analysis for aerial images**, *C*.
 Jun. 2017
 - o Trained convolutional neural network to detect human with more than 91%
 - o Analyzed human behavior with principle component analysis and vanishing point
- Nov. 2016 - **Assignment design for UIUC CS543 computer vision course**, *Matlab*.
 Jun. 2017
 - o Implemented example code and designed example architecture to train Cifar 100
 - o Wrote deep learning tutorials and assignment walkthrough instructions on Kaggle
- Jul. - Dec. 2013 **Human tracking mobile robots with Kinect**, *C++*.
 2013
 - o Identified user patterns with SIFT and GMM background subtraction algorithms
 - o Integrated depth sensor information, target user features and mobile robot control

Skills

- Languages Python, C/C++ , MATLAB, C#
- Library Pytorch, Numpy, Pandas, OpenCV, Docker, Kubernetes, AWS, Keras, Matlab, L^AT_EX

Leadership

- 2018-2023 **Summer Research Internship Program, UCSD**.
 o Mentored students in the Spring/Summer Research Intern Program (SRIP) in computer vision research.
 o **2018-2020** Collected a large-scale dataset, Objects Obtained With fLight (OOWL), with a drone. Published 3 papers using the collected dataset. *[Webpage]*
 o **2021** Collected a large-scale self-driving dataset with the explanation of the driving action. *[Webpage]*
 o **2022** Collected a large-scale visual language dataset that contains unanswerable questions. One paper accepted in ICCV23.
- 2019-2020 **GEAR Research Mentor, UCSD**.
 o Mentored students in Guided Engineering Apprenticeship in Research (GEAR) program in computer vision research.
- 2018-2019 **ENLACE bi-national summer research program, UCSD**.
 o Mentored students in ENLACE, a high school outreach program promoting diversity in research, especially in Hispanic communities.

- 2013-2014 **President of Student Association of EECS Department, NCTU.**
o Organize student events and invite speakers from the industry for tech talk

Award

- Nov 2022 NeurIPS 2022 Scholar Award
June 2022 Amazon Post-Internship Fellowship
May 2021 2021 Qualcomm Innovation Fellowship Finalist [link]
Spring 2019 UCSD graduate student association travel grant award
Jan. 2016 - Full Scholarship as exchange student at UIUC
May 2016
Jul. - Aug. 2014 Full Scholarship for an internship in Cornell University
Sep. 2012 - National Chiao Tung University scholarship
Jun. 2016

Exchange Experience

- Jan. - Jun. 2016 Exchange student at University of Illinois at Urbana-Champaign
Jul. - Aug. 2014 Short term internship in Advanced Multimedia Lab in Cornell University

Language

Mandarin, English